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DEPARTMENT OF AEROSPACE ENGINEERING

INTERPLANETARY TRAJECTORY DESIGN

JUICE Trajectory Reconstruction

SPICE-Anchored MGA-1DSM Modeling, Robust Lambert Solving, and Optimization

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Abstract

This report documents the reconstruction of the JUICE interplanetary cruise trajectory using a Sun-centered MGA-1DSM model anchored to a high-fidelity SPICE spacecraft ephemeris. The work proceeds from SPICE coverage verification and “truth” extraction, through a robust universal-variable Lambert implementation with explicit short-way/long-way branching, and concludes with constrained nonlinear optimization of DSM timing parameters. A physically meaningful flyby validation is derived from hyperbolic gravity-assist geometry, yielding turning-angle and periapsis-altitude feasibility checks. The report also records key debugging discoveries (SPK coverage clamping, branch ambiguity, solver robustness, and cost stiffness) and provides the final stabilized formulation.

1 Introduction

The JUICE mission to the Jovian system is enabled by multiple planetary gravity assists and deep-space maneuvers. A practical way to analyze such trajectories is to combine:

- **High-fidelity reference ephemerides** (SPICE SPK kernels), and
- **Reduced-order design models** (patched-conic Lambert arcs with DSMs).

This project reconstructs the trajectory using an MGA-1DSM structure: each inter-flyby leg is modeled as two heliocentric Lambert arcs separated by one DSM at an unknown time fraction $\alpha \in (0, 1)$. The DSM POSITION is anchored directly to the truth SPK, while the VELOCITY is determined by Lambert solutions and the DSM impulse.

Key Design Choice

The DSM location was anchored to SPICE truth to isolate modeling error to TIMING and VELOCITY DISCONTINUITIES, making plots visually comparable to the ephemeris and keeping optimization well-posed.

1.1 Objectives

1. Extract a consistent set of encounter epochs and truth states from SPICE.
2. Implement a robust Lambert solver (universal variables + root bracketing).
3. Resolve short-way/long-way ambiguity by evaluating both branches.
4. Optimize DSM timing parameters $\alpha_1, \dots, \alpha_5$ for a 5-leg sequence.
5. Validate flybys physically using hyperbolic turning-angle feasibility.

1.2 Trajectory Sequence and Truth Anchors

The reconstruction targets a fixed event sequence (launch and encounters) with known UTC epochs. Table 1 summarizes the anchors.

Table 1: Truth anchor events used in the reconstruction (epochs from SPICE-based analysis).

Event	NAIF ID	UTC Epoch
		Launch/Departure
399 (Earth)	2023-04-14 12:45:00 Lunar-Earth GA	399 (Earth)
2024-08-20 21:56:12 Venus GA	299 (Venus)	2025-08-31 05:28:49 Earth 2 GA
399 (Earth)	2026-09-28 11:44:55 Earth 3 GA	399 (Earth)
2029-01-17 18:34:08 Jupiter Arrival	5 (Jupiter barycenter)	<u>2031-07-21 06:30:37</u>

2 SPICE Data Handling and Coverage Clamping

2.1 Kernel Set and Frames

SPICE provides:

- Time kernels (LSK) for UTC↔ET conversions,
- Planetary constants (PCK) for radii,
- Mass constants (GM) for gravity-assist feasibility,
- SPKs for solar system bodies and the spacecraft trajectory.

States were queried in the heliocentric inertial frame **ECLIPJ2000** using `spkezr(target, et, frame, aberration, observer)` with `observer = SUN`. SPICE fundamentals are described by NAIF [1].

2.2 Coverage and the SPKINSUFFDATA Failure Mode

A common failure occurred when sampling the spacecraft state slightly outside the SPK coverage window, producing SPICE errors (e.g., **SPKINSUFFDATA**). To prevent this, the coverage interval for the spacecraft ID was computed:

$$\mathcal{C} = \text{spkcov}(\text{SPK}, \text{sclD}) \Rightarrow [t_{\min}, t_{\max}].$$

All subsequent “truth sampling” times were clamped:

$$t_{\text{in}} = \max(t_{\min}, t_{\text{event}} - \Delta t), \quad (1)$$

$$t_{\text{out}} = \min(t_{\max}, t_{\text{event}} + \Delta t). \quad (2)$$

Why Clamping Matters

Without clamping, the algorithm can request spacecraft states just outside coverage when trying to estimate inbound/outbound V_{∞} around an event, immediately terminating execution.

3 Model Formulation: Sun-Centered MGA-1DSM

3.1 Leg Structure and DSM Timing

Let event i occur at epoch t_i with heliocentric boundary positions $\mathbf{r}_i$ (planet at flyby/launch/arrival). Each leg is split by a DSM:

$$t_{\text{DSM},i} = t_i + \alpha_i (t_{i+1} - t_i), \quad \alpha_i \in (0, 1).$$

The DSM POSITION is anchored to the truth spacecraft ephemeris:

$$\mathbf{r}_{\text{DSM},i} = \mathbf{r}_{\text{truth}}(t_{\text{DSM},i}).$$

3.2 Boundary States and V_∞

From SPICE, heliocentric states are:

$$\mathbf{x}_{\text{sc}}(t) = \begin{bmatrix} \mathbf{r}_{\text{sc}}(t) \\ \mathbf{v}_{\text{sc}}(t) \end{bmatrix}, \quad \mathbf{x}_{\text{pl}}(t) = \begin{bmatrix} \mathbf{r}_{\text{pl}}(t) \\ \mathbf{v}_{\text{pl}}(t) \end{bmatrix}.$$

The hyperbolic excess velocity relative to a planet is:

$$\mathbf{v}_\infty(t) = \mathbf{v}_{\text{sc}}(t) - \mathbf{v}_{\text{pl}}(t), \quad v_\infty = \|\mathbf{v}_\infty\|. \quad (3)$$

For each flyby event, truth inbound and outbound samples were approximated by evaluating (3) at t_{in} and t_{out} (e.g. ± 2 days).

4 Lambert Problem: Universal Variable Derivation

4.1 Problem Statement

Given:

$$\mathbf{r}_1, \mathbf{r}_2, \Delta t > 0, \mu,$$

find velocities $(\mathbf{v}_1, \mathbf{v}_2)$ such that a two-body Keplerian arc connects $\mathbf{r}_1$ to $\mathbf{r}_2$ in time Δt about a central body (Sun) with gravitational parameter μ .

4.2 Transfer Geometry and the Parameter A

Let $r_1 = \|\mathbf{r}_1\|$, $r_2 = \|\mathbf{r}_2\|$ and define the transfer angle θ by:

$$\theta = 2(\|\mathbf{r}_1 \times \mathbf{r}_2\|, \mathbf{r}_1 \cdot \mathbf{r}_2), \quad (4)$$

which returns $\theta \in [0, \pi]$.

The classic universal-variable Lambert formulation uses:

$$A = \sin \theta \sqrt{\frac{r_1 r_2}{1 - \cos \theta}}. \quad (5)$$

Short-way vs Long-way

For a single-revolution transfer there are two branches:

$$\theta_{\text{short}} = \theta, \quad \theta_{\text{long}} = 2\pi - \theta.$$

A naive heuristic based on the sign of a cross-product component can appear to work for near-ecliptic cases, but the robust approach is to explicitly solve and compare both branches.

4.3 Stumpff Functions

Universal variable methods use the Stumpff functions $C(z)$ and $S(z)$:

$$C(z) = \begin{cases} \frac{1 - \cos \sqrt{z}}{z}, & z > 0, \\ \frac{\cosh \sqrt{-z} - 1}{-z}, & z < 0, \\ \frac{1}{2}, & z = 0, \end{cases} \quad S(z) = \begin{cases} \frac{\sqrt{z} - \sin \sqrt{z}}{(\sqrt{z})^3}, & z > 0, \\ \frac{\sinh \sqrt{-z} - \sqrt{-z}}{(\sqrt{-z})^3}, & z < 0, \\ \frac{1}{6}, & z = 0. \end{cases}$$

4.4 Time-of-Flight Equation

For a given z , define:

$$y(z) = r_1 + r_2 + A \frac{zS(z) - 1}{\sqrt{C(z)}}, \quad (6)$$

$$x(z) = \sqrt{\frac{y(z)}{C(z)}}. \quad (7)$$

Then the Lambert time-of-flight function is:

$$\Delta t(z) = \frac{x(z)^3 S(z) + A \sqrt{y(z)}}{\sqrt{\mu}}. \quad (8)$$

We solve for z such that:

$$F(z) \equiv \Delta t(z) - \Delta t_{\text{given}} = 0. \quad (9)$$

4.5 Robust Root Finding: Bracketing + fzero

The function $F(z)$ can be poorly behaved if intermediate quantities become non-real (e.g., $C(z) \leq 0$ or $y(z) \leq 0$), so a robust implementation:

- returns failure if invalid intermediate values occur,
- brackets a sign change by expanding an interval in z ,
- then applies **fzero** on the bracket.

Bracketing Logic

If $F(0) < 0$ (computed TOF too small), increase z until $F(z) > 0$. If $F(0) > 0$ (computed TOF too large), decrease z (negative) until $F(z) < 0$. A maximum magnitude bound prevents infinite loops.

4.6 Lagrange f, g Functions and Endpoint Velocities

Once z is found, compute:

$$f = 1 - \frac{y}{r_1}, \quad (10)$$

$$g = A\sqrt{\frac{y}{\mu}}, \quad (11)$$

$$\dot{g} = 1 - \frac{y}{r_2}. \quad (12)$$

Then:

$$\mathbf{v}_1 = \frac{\mathbf{r}_2 - f\mathbf{r}_1}{g}, \quad \mathbf{v}_2 = \frac{\dot{g}\mathbf{r}_2 - \mathbf{r}_1}{g}. \quad (13)$$

5 Objective Function and Optimization

5.1 Leg Cost Components

For leg i with DSM at $t_{\text{DSM},i}$:

- Arc 1: Lambert from $\mathbf{r}_i$ to $\mathbf{r}_{\text{DSM},i}$ giving $(\mathbf{v}_{\text{dep}}, \mathbf{v}_{\text{arr@DSM}})$.
- Arc 2: Lambert from $\mathbf{r}_{\text{DSM},i}$ to $\mathbf{r}_{i+1}$ giving $(\mathbf{v}_{\text{dep@DSM}}, \mathbf{v}_{\text{arr}})$.

The DSM impulse magnitude is the velocity discontinuity at the DSM:

$$\Delta v_{\text{DSM},i} = \|\mathbf{v}_{\text{dep@DSM}} - \mathbf{v}_{\text{arr@DSM}}\|. \quad (14)$$

The arrival V_∞ mismatch at node $i+1$ is:

$$\Delta v_{\infty, \text{arr}, i+1} = \|(\mathbf{v}_{\text{arr}} - \mathbf{v}_{\text{pl}}(t_{i+1})) - \mathbf{v}_{\infty, \text{truth}, \text{in}}(t_{i+1})\|. \quad (15)$$

To prevent PATHOLOGICAL solutions (e.g., $\alpha \rightarrow 0$ or $\alpha \rightarrow 1$), a minimum segment time constraint is imposed:

$$t_{\text{DSM},i} - t_i \geq \Delta t_{\min}, \quad t_{i+1} - t_{\text{DSM},i} \geq \Delta t_{\min}. \quad (16)$$

5.2 Total Cost

A representative total cost for 5 legs is:

$$J(\boldsymbol{\alpha}) = \sum_{i=1}^5 [\Delta v_{\text{DSM},i} + w_{i+1} \Delta v_{\infty, \text{arr}, i+1}], \quad \boldsymbol{\alpha} = [\alpha_1, \dots, \alpha_5]^T. \quad (17)$$

Weights w_{i+1} were used to control stiffness (e.g., heavier at Jupiter to enforce arrival matching).

Why “Heavy Jupiter Weight” Can Backfire

If the cost heavily penalizes Jupiter arrival mismatch without constraining the intermediate flybys (e.g., allowing arbitrary outbound V_∞), the optimizer can create UNPHYSICAL intermediate legs (huge V_∞) to satisfy Jupiter. A stable formulation also constrains outbound V_∞ at flybys or checks flyby feasibility.

5.3 Decision Variables and Constraints

The decision variables are the DSM time fractions:

$$0 < \alpha_i < 1.$$

In practice, bounds such as $\alpha_i \in [\alpha_{\min}, \alpha_{\max}]$ are used, plus the minimum segment time constraint (16).

The optimization problem is:

$$\min_{\alpha} J(\alpha) \tag{18}$$

$$\text{s.t. } \alpha_{\min} \leq \alpha_i \leq \alpha_{\max}, \quad i = 1, \dots, 5, \tag{19}$$

$$\text{Lambert arcs must be feasible (otherwise return a large penalty).} \tag{20}$$

6 Lambert Branch Selection Strategy

6.1 Failure of Simple Heuristics

Two incorrect/fragile approaches were encountered:

- **Cross-product sign heuristic:** infers long-way from a sign test on a component of $\mathbf{r}_1 \times \mathbf{r}_2$. This works only accidentally for nearly planar trajectories and fails for inclined/retrograde cases.
- **Minimum-energy selection:** chooses the branch with smaller departure speed $\|\mathbf{v}_1\|$. This can sabotage arrival matching when the “truth” path is higher energy (e.g., a more aggressive Jupiter approach).

6.2 Robust Resolution

The implemented robust standard is:

1. Solve BOTH short-way and long-way Lambert branches (when feasible).
2. Score each branch using an objective consistent with the global goal:
 - match truth velocity at the DSM time, and/or
 - match truth V_{∞} at encounter.
3. Select the branch with the lower mismatch score.

7 Flyby Validation: Turning Angle and Required Periapsis Altitude

7.1 Why “Turn Δv ” Is Not a Physical Check

A gravity assist can change the DIRECTION of $\mathbf{v}_{\infty}$ substantially. Therefore, the vector difference $\|\mathbf{v}_{\infty, \text{out}} - \mathbf{v}_{\infty, \text{in}}\|$ is not a meaningful feasibility metric by itself. The physically meaningful quantity is the TURNING ANGLE and the periapsis distance implied by hyperbolic geometry.

7.2 Turning Angle from Inbound/Outbound $\mathbf{v}_{\infty}$

Given inbound and outbound excess velocity vectors relative to the planet:

$$\mathbf{v}_{\infty, \text{in}}, \quad \mathbf{v}_{\infty, \text{out}},$$

the turning angle δ is:

$$\delta = \cos^{-1} \left(\frac{\mathbf{v}_{\infty, \text{in}} \cdot \mathbf{v}_{\infty, \text{out}}}{\|\mathbf{v}_{\infty, \text{in}}\| \|\mathbf{v}_{\infty, \text{out}}\|} \right). \tag{21}$$

7.3 Hyperbolic Orbit Relations

For a hyperbolic flyby with planet gravitational parameter μ_p and hyperbolic excess speed v_∞ , the specific orbital energy is:

$$\varepsilon = \frac{v_\infty^2}{2}.$$

The semi-major axis of the hyperbola is negative:

$$a = -\frac{\mu_p}{2\varepsilon} = -\frac{\mu_p}{v_\infty^2}.$$

Define $|a| = \mu_p/v_\infty^2$.

A standard hyperbola property relates eccentricity $e > 1$ to turning angle:

$$\delta = 2 \sin^{-1} \left(\frac{1}{e} \right). \quad (22)$$

Derivation sketch:

1. The asymptote angle (relative to the axis) satisfies $\cos \nu_\infty = -1/e$.
2. The total deflection is $\delta = 2(\pi - \nu_\infty)$.
3. Using trig identities yields $\sin(\delta/2) = 1/e$.

Thus:

$$e = \frac{1}{\sin(\delta/2)}. \quad (23)$$

The periapsis radius is:

$$r_p = |a|(e - 1) = \frac{\mu_p}{v_\infty^2} (e - 1).$$

Substituting (23):

$$r_p = \frac{\mu_p}{v_\infty^2} \left(\frac{1}{\sin(\delta/2)} - 1 \right). \quad (24)$$

Finally, required altitude above the planet radius R_p is:

$$h_{\text{req}} = r_p - R_p. \quad (25)$$

Kernel Dependency

Computing h_{req} requires both μ_p (GM) and R_p (radii). If the mass-constants kernel is missing, SPICE returns no GM and the altitude becomes NaN.

8 Implementation Overview (MATLAB + MICE)

8.1 Algorithm Summary

For each candidate vector α :

1. For each leg i :
 - (a) Compute $t_{\text{DSM},i}$ and clamp with (16).
 - (b) Query SPICE truth for $\mathbf{r}_{\text{DSM},i}$ at $t_{\text{DSM},i}$.
 - (c) Solve Lambert for Arc 1 and Arc 2, testing both short/long branches.
 - (d) If either arc fails, return a large penalty.
 - (e) Compute $\Delta v_{\text{DSM},i}$ and V_∞ mismatch at arrival.

2. Sum the costs to obtain $J(\alpha)$.

8.2 Robustness Patterns Used

- **Penalty on failure:** if Lambert returns invalid values, set $J = 10^{12}$ so the optimizer avoids that region.
- **Branch enumeration:** always attempt both branches where possible.
- **Coverage clamping:** ensure all truth-sampling times lie in SPK coverage.

8.3 Representative MATLAB Snippets

Listing 1: Clamping truth sampling to SPK coverage

```
1 t_in = max(t_cov_start, et - dt_target);
2 t_out = min(t_cov_end, et + dt_target);
```

Listing 2: Long-way handling by explicit branch choice

```
1 theta = atan2(norm(cross(r1,r2)), dot(r1,r2));
2 if longway
3     theta = 2*pi - theta;
4 end
```

Listing 3: Minimum segment-time gate to prevent alpha corner solutions

```
1 dt_min = 20*86400;
2 if (t_dsm - t0) < dt_min || (t1 - t_dsm) < dt_min
3     J = 1e12; return;
4 end
```

9 Results and Debugging Narrative

9.1 Observed Failure Modes and Fixes

1. **SPICE state extraction failures** (SPKINSUFFDATA): fixed via SPK coverage detection and clamping.
2. **Lambert NaNs / undefined objective at initial point:** fixed via bracketing + `fzero` and penalty returns on invalid geometry.
3. **Short/long-way ambiguity produced nonsensical arcs:** fixed by dual-branch solving and objective-based branch selection.
4. **Optimizer drove α to bounds and produced extreme V_∞ :** diagnosed as cost stiffness and insufficient constraints; stabilized by minimum segment-time constraints and (optionally) adding outbound V_∞ matching.
5. **NaN flyby altitude:** traced to missing GM kernel; fixed by loading mass constants (GM) kernel and using correct NAIF IDs for physical bodies.

9.2 Interpreting a “Bad Plot”

A visually distorted reconstructed trajectory (oversized loops, sharp swings, or unreasonable heliocentric excursions) typically indicates one of:

- DSM too near a node ($\alpha \approx 0$ or 1) forcing extreme velocities,
- Lambert branch mismatch (wrong short/long selection),
- cost forcing Jupiter arrival at the expense of intermediate feasibility.

Practical Diagnosis

If the printed flyby table shows v_∞ values an order of magnitude larger than expected (e.g., 20–30 km/s at Earth/Venus), the solution is not physically credible, even if a plot appears smooth.

10 Figures and Deliverables

10.1 Trajectory Plots

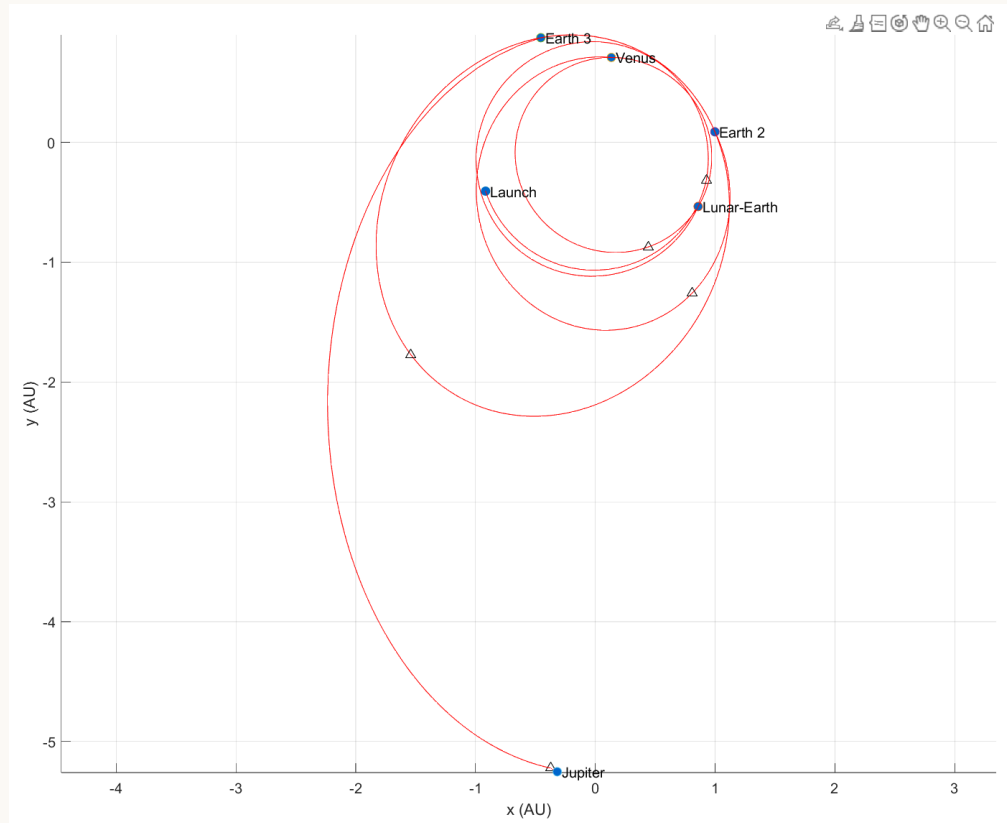


Figure 1: final reconstructed trajectory plot (truth nodes + reconstructed arcs).

10.2 Flyby Feasibility Table

Table 2: Phase B Validation: Flyby Feasibility Check

NODE (Body)	$ V_{in} $ (km/s)	$ V_{out} $ (km/s)	MagErr (km/s)	Turn (deg)	Alt_Req (km)
2 (Lunar-Earth)	3.805	3.312	0.493	80.24	10 996
3 (Venus)	7.135	7.117	0.018	42.79	5088
4 (Earth 2)	10.919	10.912	0.008	21.00	8633
5 (Earth 3)	10.887	10.968	0.080	26.53	4831

- Final Arrival (Jupiter):
 - $|V_{\infty, arr}|$: 8.573 km/s

- Mismatch vs. Truth: 2.705 km/s

11 Conclusion

This work produced a robust pipeline for reconstructing a multi-flyby interplanetary trajectory using SPICE truth anchoring and an MGA-1DSM model. The central technical insights were:

- SPICE coverage handling is mandatory for robust truth sampling.
 - A Lambert solver must treat invalid geometries safely (penalty-based optimization).
 - Short/long-way ambiguity cannot be resolved reliably by planar heuristics; both branches must be evaluated.
 - “Good-looking” plots are insufficient: physically meaningful flyby validation requires turning-angle and periapsis-altitude checks.
 - Overweighting terminal constraints can produce nonphysical intermediate legs unless flyby/outbound constraints are also enforced.
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Recommended Next Extensions

1. Add multi-revolution Lambert capability for very long legs.
2. Increase model fidelity by allowing the DSM POSITION to vary (not only time), with small bounded deviations from truth.
3. Replace single-DSM legs with a 2-DSM model on the final Earth-to-Jupiter transfer to reduce stiffness and improve Jupiter arrival matching.

References

- [1] NAIF, *SPICE Toolkit*, Navigation and Ancillary Information Facility, Jet Propulsion Laboratory, Fundamentals and Toolkit documentation, 2024.